

**MACHINE LEARNING-BASED ANALYSIS OF CHANGING
SEASONAL PATTERN IN BANGLADESH**



Submitted by

NAHID HASAN MUNNA

Roll No: MUH2006047M

Session: 2019-2020

Course Code: A.MTH-4205

*This project submitted to the Department of Applied Mathematics
in partial fulfilment of the requirements for the degree of
Bachelor of Science(Honours) in Applied Mathematics*

Department of Applied Mathematics

Noakhali Science and Technology University

Noakhali-3814, Bangladesh

Submission Date: 19 June , 2025

ACKNOWLEDGEMENT

At first, grateful to Almighty Allah for completing this project successfully. I would like to precise my heartiest thanks to my honorable teacher and supervisor, Mohammed Nizam Uddin, Associate Professor, Department of Applied Mathematics, Noakhali Science and Technology University for his inspiration and support in the several months that enable me to gain a lot of knowledge applicable to my project work.

I express my deepest sense of gratitude to all my honorable teachers of the Dept. of Applied Mathematics for their motivation, magnificent support, and manifesting suggestions in various ways for completing this project.

In the end, thanks to my parents. Without their aid and spirit throughout my life, I would not have been in a position to take a study.

Author

CERTIFICATION

This is to verify that the project work entitled “**Machine Learning-Based Analysis of Changing Seasonal Pattern in Bangladesh**”, submitted by **Nahid Hasan Munna** (MUH2006047M), to the Noakhali Science and Technology University, Noakhali – 3814, Bangladesh for the partial achievement of the requirements for the degree of Bachelor of Science in Applied Mathematics, is authentic record work carried out by her under my supervision.

Supervisor

.....

Mohammed Nizam Uddin

Associate Professor

Department of Applied Mathematics

Noakhali Science and Technology University.

Noakhali- 3814

TABLE OF CONTENTS

ACKNOWLEDGEMENT	ii
CERTIFICATION.....	iii
TABLE OF CONTENTS	iv
LIST OF FIGURES.....	v
LIST OF TABLES.....	vi
LIST OF ABBREVIATIONS	vii
ABSTRACT	viii
CHAPTER 1 INTRODUCTION.....	9
1.1 Introduction	9
1.2 Objectives of the Study	10
1.3 The terms need to be concentrated on this research.....	12
CHAPTER 2.....	14
2.1 Climate Change in South Asia – Regional Context.....	14
2.2 Role of Machine Learning in Climate Science	15
2.2.1 Limitations of Traditional Climate Models.....	15
2.2.2 Machine Learning.....	17
2.2.3 Advantages of Using Machine Learning in Climate Studies.....	18
2.2.4 Types of Machine Learning Approaches in Climate Applications.....	19
2.2.5 Used Cases of Machine Learning in Climate Science.....	22
2.2.6 Challenges and Limitations of Machine Learning in Climate Science.....	24
2.2.7 Future Directions: Integrating Machine Learning with Climate Science	25
2.3 Literature Review	27
2.4 Societal Impacts of Forecasted Climatic Changes.....	28
2.5 Limitations of the Study	32
2.6 Future Research Scope.....	33

CHAPTER 3	35
METHODOLOGY	35
3.1 Random Forests Model.....	35
3.2 Steps of Random Forest Algorithm and key concepts.....	37
Key Concepts Used:	38
3.3 Random Forest Regression Model.....	39
CHAPTER 4	40
RESULT & DISCUSSION	40
4.1 Introduction.....	40
4.2 Model Validation: Ensuring Forecast Reliability	40
Figure 4.1: High-Accuracy RF Model Validation: Actual vs. Forecasted Maximum Temperature (2016–2022)	41
4.3 Ten-Year Forecast of Climatic Variables (2023–2032)	42
4.4 Seasonal Analysis of Forecasted Climate	44
4.5 Model Comparison – Traditional vs. Machine Learning Approaches	440
CHAPTER 5.....	53
CONCLUSION	53
References	54

LIST OF FIGURES

Figure 3.1: Random Forest Algorithm.....

Figure 4.1: High-Accuracy RF Model Validation: Actual vs. Forecasted Maximum Temperature (2016–2022).....

Figure 4.2: High-Accuracy RF Model Validation: Actual vs. Forecasted Minimum Temperature (2016–2022)

Figure 4.3: High-Accuracy RF Model Validation: Actual vs. Forecasted Relative Humidity (2016–2022)

Figure 4.4: Historical (1981–2022) and 10-Year Forecast (2023–2032) of Maximum Temperature in Dhaka

Figure 4.5: Historical (1981–2022) and 10-Year Forecast (2023–2032) of Minimum Temperature in Dhaka

Figure 4.6: Historical (1981–2022) and 10-Year Forecast (2023–2032) of Relative Humidity in Dhaka.

Figures 4.7–4.11: Density and Box Plots for Forecasted Max Temp, Min Temp, and Relative Humidity by Season

LIST OF TABLES

Table 4.1: Forecasted seasonal maximum temperature from year 2023 to 2032
Table 4.2: Forecasted seasonal minimum temperature from year 2023 to 2032
Table 4.3: Forecasted seasonal RH from year 2023 to 2032
Table 4.4: Differences in forecasted seasonal maximum temperature from year 2023 to 2032
Table 4.5: Differences in forecasted seasonal minimum temperature from year 2023 to 2032
Table 4.5: Differences in forecasted seasonal RH from year 2023 to 2032

LIST OF ABBREVIATIONS

ML	Machine Learning
RFR	Random Forests Regression
RMS	Root Mean square

ABSTRACT

This study uses a machine learning-based methodology to examine how the seasonal climate patterns in Dhaka, Bangladesh, are changing. Accurate, localized climate forecasting is now more important than ever due to the growing effects of urbanization and global warming. Using historical meteorological data from 1981 to 2022, the study uses a Random Forest Regression (RFR) model to predict the highest temperature, lowest temperature, and relative humidity from 2023 to 2032. The model showed excellent predicted accuracy when tested against observed data from 2016 to 2022.

The findings indicate that while relative humidity peaks continue to be noticeable throughout the monsoon season, maximum and minimum temperatures are steadily rising, especially during the summer and winter. These results demonstrate how Dhaka's microclimate is impacted by the city's fast development and land use changes. The RFR model is more reliable at capturing nonlinear climatic variations than more conventional statistical models like ARIMA and SARIMA.

The study contributes to the emerging field of climate informatics by illustrating how data-driven environmental modeling can enhance long-term forecasting accuracy in urban contexts. The results offer valuable insights for urban planners, climate policymakers, and environmental agencies seeking to develop adaptive strategies for densely populated megacities. By leveraging artificial intelligence, this work supports proactive climate resilience planning and sustainable resource management in the face of increasing climatic uncertainties.

Keywords: Climate change, Random Forest Regression, Time Series Forecasting, Seasonal Variability.

CHAPTER 1

INTRODUCTION

1.1 Introduction

Global warming means the increase of the mean temperature of the Earth's surroundings. People think that global warming and climate change are the same. In the past 100 years, the average temperature of air in earth surface has risen by approximately 1-degree Celsius. The increase in earth temperature plays an important role in global warming. Besides the rise in CO₂, fossil fuel burning, cutting trees, toxic gases from air conditioners and refrigerators are also responsible for global warming. Therefore these absorb the sunlight, and the earth surfaces are being heated and as a result people have to face many difficulties with high temperature. Some activities of human are also responsible for climate change and global warming like using motor vehicles, refrigerators, cutting trees filling up rivers and canals [1]. Due to climate change the Asian winter monsoon has extincted because the snow cover is decreasing day by day. Now it can be said that Asia would face rising trends in mean surface temperature (0.25 to 0.34 degree Celsius per decade). The intergovernmental panel on climate change (IPCC) developed by the world meteorological organization and UNEP to know about climate change and global warming because people are mostly responsible for this. [2]

Now it is obvious that our main concern have to be the change in temperature, climate change, global warming. In south Asian countries, Bangladesh is one of the most overpopulated countries and as Dhaka is the capital of Bangladesh, most of the people from village and urban area come to Dhaka for job purpose and other facilities and the harmful activities they are doing here causes global warming and plays impact on seasonal temperature change. To forecast the upcoming temperature for every season like summer, monsoon, autumn and winter in Dhaka, Bangladesh we are to use the machine learning based Random Forest algorithms. Machine learning techniques especially Random Forest model can provide reasonable weather predictions without any precise idea on atmospheric physics. This can play a vital role to enhance forecasting capabilities

[3].

While previous studies have explored temperature forecasting in Bangladesh using conventional statistical methods such as ARIMA or SARIMA, these models often fall short in handling complex, nonlinear climate data. Moreover, most existing research focuses on short-term forecasts or nationwide averages, lacking city-specific insights. Few studies have applied advanced machine learning methods to predict seasonal climate variability in Dhaka, leaving a significant gap in high-resolution, data-driven forecasting for urban climate adaptation. Many scientists have discovered that automated tools and apparatus should be included in real-time forecasting systems. Therefore, one method that works well for many past weather prediction studies is the Random Forest algorithms [4].

The machine learning based random forest (RM) algorithm works by combining many small models for making perfect prediction. This model also maintains the principles of statistical studies and is used in both classification and regression-based tasks. The applied areas of this model are very wide such as risk analysis, remote Sensing and flood prediction. Random forest algorithms mainly work perfectly with large and complex data and one key feature of it is to rank the significance of each variable. As this model has these strengths, this model is suitable for handling non-linear and multi variable problems and the temperature prediction is one of them. It also improves the accurate prediction for seasonal temperature changes [5] .

This research is significant for the accuracy of temperature forecasts in the capital of Bangladesh (Dhaka). It will take decisions smartly in resource management and temperature adaptation that will help to predict accurate temperature. To fill up the present gaps and limitations, this project will measure maximum, minimum temperature and also the relative humidity (RH) in Dhaka city for the period between 2025 and 2032. The main goal of this project is to provide fully accurate and location specific temperature forecasts using the previous large weather datasets and model (RF).

The study does not include other climatic factors such as wind speed, rainfall, or solar radiation due to data limitations. Spatial heterogeneity within Dhaka is also not captured, as the analysis is based on city-wide averages. While RFR provides strong predictive power, it lacks physical interpretability compared to physics-based models.

1.2 Objectives of the Study

This research aims to:

1. Develop robust forecasts of daily maximum and minimum temperatures alongside relative humidity in Dhaka (2023–2032), leveraging four decades (1981–2022) of climatic data through advanced Random Forest Regression techniques.
2. Decipher seasonal and inter-annual variability across Winter, Summer, Monsoon, and Autumn, elucidating the nuanced impacts of climate change on local atmospheric dynamics.
3. Benchmark the predictive efficacy of Random Forest against classical statistical models to establish superior methodologies for urban climate time series forecasting.
4. Elucidate the influence of rapid urbanization and land-use transformation on microclimatic trends, integrating forecast outcomes with urban environmental processes.
5. Generate empirically grounded recommendations for resilient urban planning and public health policy, addressing emergent climatic uncertainties in densely populated South Asian megacities.

1.3 The terms need to be concentrated on this research

Time Series: A time series is a sequence of data points collected or recorded at successive points in time, usually at equally spaced intervals (e.g., daily, monthly, yearly). Example: Daily stock prices, monthly rainfall, annual GDP.

Time series analysis: It refers to the techniques used to understand the underlying patterns and structure of time series data, and to make forecasts or informed decisions based on it.

Trend: Trend refers to the long-term movement or direction in a time series. It shows the general tendency of the data to increase, decrease, or remain constant over an extended period. Trends are often influenced by structural factors like economic growth, technological advancement, or demographic changes. Identifying the trend helps to understand the overall behavior of the data over time.

Seasonality: Seasonality is the repeating pattern or fluctuation in a time series that occurs at regular intervals, such as daily, monthly, or yearly. These patterns are driven by seasonal factors like climate, holidays, or recurring social activities. For example, retail sales often increase during the festive season, or electricity consumption may rise during summer due to air conditioning.

Cyclic pattern: It refers to fluctuations in a time series that occur over longer periods and are not fixed like seasonal patterns. These cycles are often related to economic or business cycles and may span several years. Unlike seasonality, cyclic variations do not follow a strict calendar-based repetition and are usually influenced by external macroeconomic factors such as inflation, interest rates, or political events.

Irregular variation: Irregular variation also known as random or residual variation, represents unpredictable and random changes in a time series. These are caused by unforeseen events such as natural disasters, strikes, or accidents. Irregular variations do not follow any pattern or trend and are considered noise in the data, often removed or minimized during analysis to reveal the underlying structure.

RH: RH stands for Relative Humidity. Relative Humidity (RH) is the percentage of water vapor present in the air compared to the maximum amount of water vapor the air can hold

at a given temperature. It indicates how "humid" or "moist" the air feels .When RH is 100%, the air is fully saturated with moisture and cannot hold any more water vapor, leading to do, fog, or precipitation. When RH is low, the air feels dry and evaporation occurs more quickly. Formula for RH:

$$RH = (\text{Actual Vapor Pressure} \div \text{Saturation Vapor Pressure}) \times 100\%$$

Mean (Average): The sum of all values divided by the number of values. In temperature forecasting, it often refers to the mean temperature over a specific period.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Residual: The difference between observed and predicted values. In forecasting, analyzing residuals helps to improve model accuracy.

$$e_i = y_i - \hat{y}_i$$

Regression: A statistical method used to model the relationship between a dependent variable (e.g., temperature) and one or more independent variables (e.g., humidity, wind speed).

$$y_t = \beta_0 + \beta_1 x + \epsilon$$

RMSE (Root Mean Square Error): A commonly used measure of error in forecasts. It gives the average magnitude of prediction error in the same units as the data.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

CHAPTER 2

LITERATURE REVIEW

2.1 Climate Change in South Asia – Regional Context

South Asia, home to nearly one-fourth of the world's population, is among the most climate-vulnerable regions globally due to its unique geography, socio-economic challenges, and high population density. The region has seen a marked increase in the frequency and severity of extreme weather events, including heatwaves, unseasonal rainfall, floods, and cyclones, which pose significant threats to human livelihoods, ecosystems, and economies [6].

In Bangladesh, one of the most climate-sensitive countries, mean surface temperatures have risen by approximately 0.5°C since the 1970s, with projections indicating further increases in coming decades [7]. The IPCC Sixth Assessment Report projects temperature increases in South Asia ranging between 1.5°C and 2°C by the end of the 21st century under moderate greenhouse gas emission scenarios. These rising temperatures are expected to disrupt seasonal weather patterns, affecting agricultural cycles, freshwater availability, and urban living conditions [8].

The South Asian monsoon, vital for the agrarian economies of the region, is becoming increasingly erratic, characterized by delays in onset, spatial variability, and shorter durations [9]. Such monsoonal irregularities threaten crop yields and food security, contributing to increased rural distress and migration to urban centers [10]. Coastal Bangladesh faces saltwater intrusion caused by rising sea levels, reducing the fertility of agricultural lands and endangering local food production systems [11].

The melting of Himalayan glaciers presents an additional regional challenge. These glaciers feed major rivers such as the Ganges, Brahmaputra, and Indus, which support hundreds of millions of people. Accelerated glacial retreat threatens long-term water security and raises the risk of glacial lake outburst floods (GLOFs), which can cause devastating downstream flooding [12].

Urban heat island (UHI) effects exacerbate warming trends in major metropolitan areas including Dhaka, Delhi, and Karachi. Rapid urbanization, loss of green spaces, increased vehicular emissions, and unplanned construction have intensified daytime and nighttime temperatures in these cities [10]. This amplification of heat poses serious public health risks, including heatstroke, respiratory ailments, and increased incidence of vector-borne diseases such as dengue and malaria [13].

Moreover, the Bay of Bengal is witnessing an increase in the frequency and intensity of tropical cyclones. Events like Cyclone Amphan in 2020 and Cyclone Mocha in 2023 have caused widespread displacement, infrastructure damage, and significant economic loss. These cyclones highlight the compound risks faced by coastal populations due to climate change.

Given these pressing challenges, there is an urgent need to develop precise climate forecasting tools and resilient adaptation policies. Machine learning approaches that forecast temperature and humidity trends, as explored in this study, offer valuable support for proactive climate risk management. Understanding the regional context of South Asia’s climate variability is essential for interpreting localized projections—such as those for Dhaka—within the broader environmental and socio-economic landscape.

2.2 Role of Machine Learning in Climate Science

Climate science, by its nature, deals with large-scale, high-dimensional, and complex datasets involving interactions between atmospheric, oceanic, and terrestrial processes. Traditional analytical methods, although powerful, often fall short when attempting to capture the intricate, nonlinear, and rapidly changing patterns seen in today’s climate data. In this context, machine learning (ML) has emerged as a transformative tool that can handle large datasets, learn from patterns, and produce accurate forecasts, thereby enhancing climate modeling and environmental monitoring.

2.2.1 Limitations of Traditional Climate Models

Traditional climate modeling techniques, such as General Circulation Models (GCMs),

Autoregressive Integrated Moving Average (ARIMA), and linear regression, have historically been essential tools in climate science. However, these models exhibit significant limitations that reduce their effectiveness for modern forecasting requirements, particularly at local and regional scales, including city-level temperature predictions [14].

One of the primary challenges associated with GCMs is their substantial computational demand. GCMs simulate Earth's atmosphere, oceans, and land surface interactions through complex mathematical formulations based on physical laws. While invaluable for global and long-term climate projections, running high-resolution GCM simulations requires extensive computing power, often involving supercomputers and multi-day processing times. This computational intensity makes GCMs impractical for real-time forecasting or for applications in resource-limited settings such as Bangladesh, where timely, localized climate information is crucial for adaptation and disaster preparedness [6].

Statistical models like ARIMA and linear regression face additional drawbacks stemming from their inherent assumptions. These models generally rely on assumptions such as linearity, stationarity, and normality of data distributions. Climate variables, however, frequently display nonlinear behaviors, abrupt regime shifts, and pronounced seasonality that violate these assumptions. For example, temperature and humidity data often exhibit complex temporal dynamics and extreme events that standard linear models cannot accurately capture. When these assumptions are breached, the predictive accuracy of such models diminishes considerably, limiting their utility for climate applications [9].

Another critical limitation of traditional models is their sensitivity to data quality issues. Long-term climate datasets, especially from developing regions, often contain missing entries, measurement errors, and inconsistencies due to equipment malfunctions or reporting gaps. Conventional modeling techniques generally require complete and clean datasets for reliable functioning, and their inability to effectively handle missing or noisy data can lead to biased or unreliable forecasts [15].

Furthermore, these models lack flexibility and scalability when confronted with rapidly changing environmental conditions. Traditional climate models often need manual recalibration when applied to new locations or changing temporal contexts, limiting their adaptability. This rigidity poses challenges in rapidly urbanizing areas such as Dhaka,

where evolving factors like land use changes, air pollution, and population growth strongly influence local climate dynamics [10]. In contrast, machine learning approaches offer greater adaptability by dynamically learning from large, diverse datasets and continuously updating models to reflect recent patterns and variables [16].

While traditional climate models have significantly advanced our understanding of atmospheric processes, their limitations in computational efficiency, rigid assumptions, data quality sensitivity, and inflexibility restrict their applicability for high-resolution, near-real-time forecasting. These challenges underscore the necessity for more adaptive and robust modeling approaches—such as machine learning—that can better accommodate the complexity and variability of contemporary climate systems.

2.2.2 Machine Learning

A sub-field of artificial intelligence called machine learning (ML) allows systems to learn from data and enhance their functionality without explicit programming. Applications for it are numerous and include speech processing, picture identification, and predictive analysis. Supervised, unsupervised, and reinforcement learning are the three primary categories of machine learning models. Using labeled data to train a model to predict or categorize results is known as supervised learning. Unsupervised learning, on the other hand, works with unlabeled data and looks for hidden patterns or groups in the data. The foundation of reinforcement learning is the idea that an agent learns from interactions with its surroundings by being rewarded or punished for its actions. For applications like sentiment analysis, spam detection, and forecasting, supervised learning is frequently utilized. Pattern identification, anomaly detection, and customer segmentation are common applications of unsupervised learning. Applications for reinforcement learning include self-driving cars, gaming, and robotics. The kind of data available and the nature of the problem determine which machine learning type is used.

2.2.3 Advantages of Using Machine Learning in Climate Studies

ML models offer several advantages over traditional approaches:

Nonlinear Modeling Capability

One of the key advantages of machine learning (ML) in climate studies is its ability to capture complex nonlinear relationships. Unlike traditional linear models, ML techniques can identify and learn intricate patterns between multiple input variables, such as temperature, humidity, and pressure, and the target outcomes, like future temperature or precipitation levels. This capability is particularly valuable for atmospheric phenomena that often do not follow simple or predictable trends, allowing for more accurate and nuanced forecasting.

Ability to Handle Large Datasets

Climate science generates vast amounts of data from sources such as decades of historical weather records, satellite observations, and detailed time series measurements. ML algorithms, including Random Forests, Gradient Boosting Machines, and Neural Networks, are well-equipped to process and learn from these large and complex datasets. Their capacity to manage high-dimensional data enables scientists to extract meaningful insights and improve prediction accuracy without the need for extensive data preprocessing.

Missing Data Tolerance

Another advantage of many ML approaches is their robustness in handling incomplete or noisy data, which is common in climate datasets due to sensor failures or gaps in recording. ML methods can incorporate imputation techniques to estimate missing values or adapt their learning process to work around these gaps. This flexibility ensures that the models remain reliable and effective even when the data is imperfect.

Feature Selection and Interpretation

Some machine learning algorithms, such as Random Forests, offer additional benefits by providing measures of feature importance. These rankings help researchers identify which

variables have the most significant impact on the model's predictions. This interpretability not only aids in understanding the underlying climate processes but also assists in refining models and guiding further data collection efforts.

Fast Computation and Real-Time Prediction

Once trained, ML models can generate predictions rapidly, often within seconds. This speed enables their use in real-time applications, such as issuing timely flood warnings or urban heat alerts, which are critical for disaster preparedness and response. The computational efficiency of ML models makes them highly practical for continuous monitoring and forecasting in dynamic environmental conditions.

Adaptability and Continuous Learning

Machine learning models are inherently adaptable and can be retrained or updated as new climate data becomes available. This continuous learning capability allows them to adjust to evolving environmental patterns, improving their performance over time. Such flexibility is essential in a changing climate where historical trends may no longer fully represent future conditions.

2.2.4 Types of Machine Learning Approaches in Climate Applications

A. Supervised Learning

Supervised learning is one of the most widely used machine learning approaches in climate science. It involves training models on labeled datasets, where the input features are paired with known target outcomes. The goal is for the model to learn the underlying relationship between inputs and outputs, so it can accurately predict future values or classify new data points. For example, input features could include historical temperature, humidity, and atmospheric pressure, while the targets might be future temperature values or rainfall amounts.

Common algorithms in supervised learning include Linear and Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), and Neural Networks. Linear regression can model simple relationships between variables, while decision trees and random forests capture nonlinear interactions and variable importance, making them

powerful for complex climate data. Neural networks, especially deep learning models, excel in handling large datasets with multiple features, enabling them to capture intricate spatial and temporal climate patterns.

In practical climate applications, supervised learning is extensively used for forecasting key weather variables such as temperature, precipitation, and wind speed. These predictions are crucial for agriculture, disaster preparedness, and energy management. For example, predicting heatwaves or drought events helps governments and communities prepare in advance, mitigating adverse impacts on health, agriculture, and water resources. Similarly, estimating crop yields under changing climate conditions allows farmers and policymakers to make informed decisions about crop selection and resource allocation.

Moreover, supervised learning models can be continually updated with new data, improving their accuracy over time and adapting to climate change dynamics. Their interpretability, especially in tree-based models, helps researchers understand which climatic factors most influence predictions. Despite their strengths, supervised models require large amounts of labeled data, which can be a limitation in some regions or for rare extreme events. Nonetheless, their predictive power and versatility make supervised learning a fundamental tool in climate research and operational forecasting.

B. Unsupervised Learning

Unsupervised learning is a powerful approach when dealing with unlabeled climate data, where the goal is to uncover hidden patterns, structures, or relationships without predefined output labels. This makes it especially useful in exploratory data analysis, where scientists want to identify natural groupings or anomalies within large and complex climate datasets.

Key algorithms in unsupervised learning include K-Means Clustering, Principal Component Analysis (PCA), and Autoencoders. K-Means clustering partitions data into groups based on feature similarity, which can help classify climatic zones or weather regimes without prior labeling. PCA is a dimensionality reduction technique that transforms high-dimensional data into a smaller number of uncorrelated variables called principal components, which capture most of the variability in the data. This

simplification aids in visualization and noise reduction. Autoencoders, a type of neural network, learn compact representations of data and are effective for anomaly detection.

Applications of unsupervised learning in climate science are diverse. One common use is classifying climatic zones based on observed variables such as temperature and precipitation patterns. This classification helps in understanding regional climate variability and guiding local adaptation strategies. Unsupervised methods are also instrumental in detecting abnormal climate events, such as sudden temperature spikes or shifts in weather patterns, which may be early indicators of extreme phenomena or climate change.

Dimensionality reduction techniques like PCA enable researchers to visualize complex climate data more easily and identify dominant modes of variability, such as El Niño Southern Oscillation (ENSO) or monsoon cycles. By highlighting the key factors driving climate variability, unsupervised learning facilitates the development of better models and informs policy decisions.

Unsupervised learning does not require labeled data, which is advantageous in many climate applications where comprehensive labeling is unavailable or costly. However, interpreting the results often requires domain expertise, as the discovered patterns may not always correspond to well-known climate processes. Despite this, unsupervised learning remains an essential tool for advancing our understanding of climate systems and their underlying complexity.

C. Reinforcement Learning

Reinforcement learning (RL) is an advanced machine learning paradigm where an algorithm learns to make decisions by interacting with an environment and receiving feedback in the form of rewards or penalties. Unlike supervised and unsupervised learning, RL focuses on learning optimal strategies or policies to maximize long-term outcomes, making it well-suited for dynamic and sequential decision-making problems.

In climate and environmental applications, reinforcement learning is increasingly being explored for tasks requiring adaptive control and optimization under uncertainty. One prominent example is adaptive weather routing for aircraft. By continuously assessing

real-time weather data and adjusting flight paths, RL algorithms can optimize routes to reduce fuel consumption, avoid turbulence, and improve safety. This application demonstrates RL's ability to handle complex, time-dependent environmental inputs and produce actionable strategies.

Another area where RL is valuable is in real-time control systems for climate-based automation. For instance, smart energy management systems in buildings or cities can use RL to optimize heating, ventilation, and air conditioning (HVAC) operations, balancing energy efficiency with occupant comfort under changing weather conditions. Similarly, RL can be applied to irrigation scheduling in agriculture, dynamically adjusting water use based on soil moisture, weather forecasts, and crop needs.

Moreover, reinforcement learning plays a role in simulation-based environmental policy modeling. Policymakers can use RL-driven simulations to explore the consequences of various interventions—such as carbon taxes, emissions regulations, or conservation incentives—and identify strategies that optimize environmental and economic outcomes. By iteratively learning from simulated feedback, RL can help design adaptive policies that remain effective amid evolving climate conditions.

While RL holds great promise, its application in climate science is still emerging and faces challenges such as the need for accurate environment models and substantial computational resources. Nonetheless, its ability to learn from interaction and optimize sequential decisions makes reinforcement learning a powerful tool for tackling complex, dynamic problems in climate adaptation and mitigation.

2.2.5 Used Cases of Machine Learning in Climate Science

Machine learning (ML) has emerged as a powerful tool in climate science, offering new capabilities to analyze complex environmental data, improve forecasting accuracy, and enhance decision-making for climate adaptation and mitigation. Its applications span a broad range of climate-related challenges, demonstrating significant impact in recent years.

One of the most prominent applications is climate forecasting, where ML algorithms predict key weather variables such as temperature, rainfall, humidity, and wind speed at various temporal scales—daily, monthly, or seasonal. Models like Random Forests (RF) and Artificial Neural Networks (ANN) have been successfully applied to forecast climatic parameters in urban centers vulnerable to climate change. For example, studies in South Asian megacities including Dhaka, Delhi, and Karachi have shown that these ML models outperform traditional statistical methods in capturing local climate dynamics and variability [17]. Improved forecasting aids by urban planners and emergency services in managing heatwaves, floods, and other extreme weather events.

Another critical case is in flood prediction and early warning systems. Flooding is a recurrent and devastating hazard in many regions, particularly in South Asia and Southeast Asia. ML models integrate diverse datasets—such as rainfall intensity, soil moisture, river discharge, and topography—to estimate flood risk in real-time. For instance, ML-driven flood warning systems deployed in India and Bangladesh use Random Forest and Gradient Boosting algorithms to analyze hydrological data streams, offering early alerts that help mitigate human and economic losses [18]. The ability of ML to process heterogeneous data sources and detect subtle patterns significantly improves the lead time and accuracy of flood predictions.

Air quality prediction is another vital application where ML excels. Pollution forecasting requires assimilating data from multiple pollutant measurements—PM2.5, nitrogen oxides (NO_x), sulfur dioxide—and meteorological variables. ML models, including Support Vector Machines (SVM) and deep learning networks, have been trained to predict air quality indices in urban areas. This information is essential for issuing health advisories and designing pollution control policies, particularly in rapidly urbanizing regions facing severe air quality challenges.

In the agricultural sector, ML supports risk management by forecasting droughts and optimizing crop planting decisions. Drought forecasting models use historical climate data combined with soil moisture and vegetation indices to predict water stress periods. These models assist farmers in making timely decisions about crop selection and irrigation scheduling, thereby enhancing resilience to climate variability. Additionally, ML helps estimate crop yields under varying climate scenarios, which is crucial for food

security planning.

Machine learning also facilitates land use and land cover (LULC) change detection through the analysis of satellite imagery. Deep learning models, such as convolutional neural networks (CNNs), are adept at identifying patterns of deforestation, urban sprawl, wetland loss, and other environmental changes over time. These techniques provide high-resolution, up-to-date monitoring, supporting conservation efforts and sustainable land management. Furthermore, ML techniques are applied to classify extreme climate events. Neural networks and SVMs help distinguish and categorize phenomena like cyclones, heatwaves, and cold spells based on meteorological data. Accurate classification aids early warning systems and improves understanding of extreme event frequency and intensity trends in a changing climate [19].

Lastly, ML contributes significantly to sea-level rise and glacier monitoring. By integrating remote sensing data with climate projections, ML models track glacier retreat and predict future sea-level changes. This capability is essential for coastal risk assessment and adaptation planning.

In summary, machine learning's ability to analyze large, complex datasets and learn nonlinear relationships makes it invaluable in addressing climate challenges. Its applications—from forecasting to environmental monitoring—enhance predictive accuracy, provide actionable insights, and support timely decision-making in climate science.

2.2.6 Challenges and Limitations of Machine Learning in Climate Science

Machine learning (ML) has emerged as a transformative tool in climate science, enabling researchers to analyze vast datasets and improve forecasting accuracy. However, despite its advantages, ML faces several inherent challenges that limit its full potential in climate applications.

A fundamental challenge lies in dependency **on data quality**. ML algorithms require large amounts of accurate and representative data to learn effectively. Climate datasets, especially those from developing regions, often contain missing values, measurement errors, and biases due to limited monitoring infrastructure or inconsistent data collection

practices [16]. Incomplete or biased data can significantly degrade model performance, leading to unreliable predictions or overconfident results. For example, satellite data may have spatial or temporal gaps, and ground-based sensors can suffer from calibration drift or failure, complicating the training process.

Another major limitation is the **lack of interpretability** in many ML models, particularly deep learning methods. These models are often referred to as "black boxes" because they generate predictions without providing clear explanations of how input features influence outputs [20]. This opacity hinders trust among climate scientists and policymakers, who require transparent models to justify decisions and implement effective mitigation or adaptation strategies. Without interpretability, ML outputs risk being dismissed or underutilized despite their potential accuracy.

Overfitting is a persistent concern in ML applications. This occurs when a model fits the training data too closely, including noise and outliers, resulting in poor generalization to new data. Overfitting is especially problematic in climate science due to the limited length of high-quality historical records or when modeling rare extreme events. Ensuring robust validation techniques and regularization methods is essential but challenging.

From a technical perspective, ML models, especially those involving deep neural networks, demand significant **computational resources** such as GPUs and large memory capacities. High-resolution climate datasets spanning multiple variables and long time periods require intensive processing power that may be unavailable in many research institutes or developing countries. This restricts the accessibility and scalability of ML approaches globally.

Lastly, ML models typically lack **physical grounding**. Unlike traditional climate models built on well-understood physical laws (e.g., conservation of energy, fluid dynamics), ML algorithms learn patterns purely from data without inherent knowledge of the underlying atmospheric or thermodynamic processes [16]. This can lead to physically inconsistent or unrealistic predictions unless domain constraints are explicitly incorporated, limiting scientific trustworthiness and the ability to extrapolate beyond observed conditions.

2.2.7 Future Directions: Integrating Machine Learning with Climate Science

To address these limitations and advance the use of ML in climate science, several promising research avenues are being pursued.

One key development is the creation of **hybrid models** that blend physics-based simulations with machine learning techniques. By embedding physical constraints and conservation laws within ML frameworks, hybrid models aim to improve predictive accuracy while maintaining interpretability and adherence to scientific principles. For example, combining General Circulation Models (GCMs) with neural networks can enhance regional climate projections by correcting biases and learning unresolved processes.

Transfer learning represents another important frontier, where models trained on large, data-rich regions are adapted for use in data-scarce locations. This approach reduces the need for extensive local training data, enabling climate predictions in under-monitored regions such as parts of South Asia or Africa. Transfer learning leverages shared patterns across geographies while fine-tuning for local specificities, making ML more scalable globally.

The field of **Explainable AI (XAI)** is rapidly evolving to make ML models more transparent and interpretable. Techniques such as feature importance scoring, saliency maps, and surrogate models provide insights into model decision-making processes. For climate science, XAI can build confidence among stakeholders by elucidating which variables drive forecasts and how predictions respond to changing inputs.

Additionally, advances in **real-time environmental sensing** via the Internet of Things (IoT) are creating new opportunities for ML-powered monitoring and early warning systems. Distributed sensor networks collect continuous data on temperature, humidity, precipitation, and pollutants, which ML models analyze instantaneously to detect hazards such as flash floods or heatwaves. These systems enable proactive disaster response and adaptive resource management, particularly in vulnerable urban and rural areas.

Overall, the integration of ML with physical modeling, transfer learning, explainable AI, and IoT-driven real-time monitoring holds great promise to overcome current challenges. This multidisciplinary approach will enhance the robustness, accessibility, and practical utility of ML in tackling complex climate problems, supporting global efforts to understand and mitigate climate change impacts.

2.3 Literature Review

Random Forest algorithms have proven effective in forecasting severe weather by learning complex atmospheric relationships. [4] demonstrated RF's capability to produce calibrated, probabilistic forecasts using variables like temperature and relative humidity. This model's adaptability makes it suitable for analyzing seasonal temperature and RH patterns in Bangladesh. [3] demonstrated that Random Forest models outperform traditional techniques in predicting weather parameters like temperature and humidity using minimal data. This method's robustness in seasonal pattern analysis suggests its high applicability for forecasting maximum and minimum temperatures and RH in climate-sensitive regions like Bangladesh. [2] applied ensemble learning, including Random Forest, to forecast temperature patterns in Bangladesh using 20 years of data. Their results showed Random Forest performed best during training, especially for maximum and minimum temperature prediction, making it a strong model for analyzing seasonal temperature and humidity trends in tropical regions like Bangladesh.

[1] analyzed a century of Bangladesh's temperature data using regression and SVR models to predict future trends. Although Random Forest wasn't used, their study underscores the feasibility of machine learning in forecasting maximum, minimum, and average temperatures, highlighting the potential for more adaptive models like RF to capture complex seasonal patterns and RH variability. [21] applied the SARIMA model to forecast Bangladesh's maximum temperature from 1981 to 2042, revealing a consistent rising trend. Though effective in capturing seasonality, the study highlights the need for more adaptive models like Random Forest, which can better handle nonlinear relationships in temperature and RH variations across seasons. [22] used Random Forest and Boruta algorithms to identify key weather variables like temperature and RH in forecasting drought across Bangladesh. Their study confirmed RF's strength in modeling seasonal climate variation and emphasized its role in predicting spatio-temporal changes in maximum/minimum temperature and relative humidity patterns effectively. [23] enhanced climate model precision by applying Random Forest with seasonal bias correction, improving predictions of maximum and minimum temperatures. Their model outperformed others across multiple climate scenarios, showing strong reliability. This reinforces RF's effectiveness in capturing seasonal temperature and humidity patterns—

essential for climate-sensitive regions like Bangladesh. [24] compared various machine learning models and found Random Forest to be effective in predicting daily and monthly temperature and humidity. Its high accuracy and reliability across time scales underscore RF's value in modeling seasonal climate patterns, making it suitable for applications in humid tropical regions like Bangladesh

[25] applied Random Forest and ensemble models to predict rainfall and daily average temperature in Bangladesh using a decade of weather data. Their findings show RF's precision in modeling temperature and humidity patterns, affirming its suitability for analyzing changing seasonal trends in maximum/minimum temperature and RH in climate-sensitive regions

[26] developed a data-driven method using Random Forest to generate high-resolution temperature maps across Bangladesh. Their approach accurately captured seasonal variations using sparse observational data, highlighting RF's effectiveness for analyzing temperature trends in developing regions—making it highly applicable for seasonal pattern analysis involving maximum, minimum temperatures and RH in Bangladesh

Incorporating humidity, wind, and precipitation into predictive models, the study confirmed Random Forest's superior performance among machine learning approaches for accurate temperature forecasting. Its strong generalization ability and stability across varying conditions make RF ideal for capturing seasonal climate variability in sensitive regions like Bangladesh, as concluded by [27]. This study employed Random Forest among other models to assess climate-sensitive disease risks using temperature and RH variables. Although LSTM performed better, RF effectively identified humidity and temperature as key seasonal predictors, validating its application for climate-based pattern analysis in tropical regions like Bangladesh, as shown by [28].

2.4 Societal Impacts of Forecasted Climatic Changes

The projected climatic changes in Dhaka, particularly the rise in seasonal temperatures and the fluctuations in relative humidity, pose significant challenges to the city's sustainable development. As one of the most densely populated cities in the world, Dhaka is especially vulnerable to even slight alterations in climate patterns. These changes can

impact everything from daily life to long-term planning in sectors such as urban development, public health, energy consumption, and water management. Understanding these effects is crucial for creating adaptive strategies that ensure the city's resilience and well-being.

Urban Planning and Infrastructure:

Urban areas, particularly megacities like Dhaka, are prone to the urban heat island (UHI) effect, where concrete structures, traffic emissions, and industrial activity collectively cause city temperatures to rise significantly higher than surrounding rural areas. The forecasted increase in maximum temperatures will likely intensify this phenomenon. As a result, building materials may degrade faster, road surfaces may become more susceptible to damage, and urban energy consumption for cooling will increase dramatically. Urban planners will need to consider climate resilience in the design and development of infrastructure. For instance, incorporating green roofs, vertical gardens, and heat-reflective surfaces can help mitigate extreme heat. The creation of more shaded public spaces, especially in high-density residential and commercial zones, will be vital in reducing heat exposure and enhancing urban livability.

Public Health:

Climatic changes, particularly rising temperatures and varying humidity levels, have profound effects on public health. Heatwaves can cause heat exhaustion, dehydration, and heatstroke, especially among vulnerable populations such as the elderly, children, pregnant women, and individuals with pre-existing health conditions. Additionally, warmer and more humid conditions are conducive to the proliferation of disease vectors like mosquitoes. Consequently, diseases such as dengue fever, chikungunya, and malaria may become more frequent and widespread. Furthermore, high humidity combined with air pollution can exacerbate respiratory conditions, such as asthma and bronchitis. The health sector must therefore be prepared with enhanced emergency response mechanisms, climate-sensitive disease surveillance, and accessible healthcare facilities, particularly in underserved urban areas.

Energy Consumption:

Temperature increases will naturally drive-up energy consumption, primarily due to increased reliance on air conditioning, fans, and refrigeration systems. In summer and monsoon months, the electricity demand may outpace supply, resulting in frequent power outages. This imbalance can disrupt daily life, especially in hospitals, schools, and businesses, leading to broader socio-economic consequences. To mitigate this, government and private sectors must invest in energy-efficient technologies and promote renewable energy sources such as solar power. Moreover, incentives for using energy-efficient appliances and smart grid technologies can help reduce the load on existing power infrastructure. Climate-responsive architectural design, including better insulation and natural ventilation, should also be promoted to reduce dependency on mechanical cooling systems.

Water Management:

Changes in temperature and humidity significantly influence the hydrological cycle. Increased evaporation rates can lead to a decline in water availability, affecting both surface and groundwater sources. Conversely, intense and erratic rainfall patterns can lead to frequent urban flooding and waterlogging, particularly in low-lying areas of Dhaka. The city's drainage systems, already overburdened, may become even more strained, leading to prolonged inundation, property damage, and waterborne diseases. Addressing these challenges requires comprehensive urban water management plans that include upgrading drainage infrastructure, constructing rainwater harvesting systems, and implementing early warning mechanisms. Additionally, integrating climate models into urban planning can help predict and prepare for extreme weather events.

Food Security and Agriculture:

Though Dhaka itself is not a major agricultural hub, its population depends heavily on the agricultural productivity of surrounding regions. Forecasted climatic changes, including heat stress and altered rainfall patterns, could reduce crop yields and disrupt supply chains. This would lead to food price volatility and increased malnutrition risks among poor urban populations. Urban food security strategies such as promoting rooftop gardening, vertical farming, and local food markets could provide partial relief. Government support in stabilizing food supply chains and developing heat-resistant crop

varieties is also crucial in ensuring long-term food security.

Migration and Housing:

Climate change-induced displacement is becoming a reality for many coastal and riverine communities in Bangladesh. As environmental degradation and natural disasters render certain areas uninhabitable, Dhaka continues to experience an influx of climate migrants. This population growth puts immense pressure on the already overburdened housing, sanitation, and healthcare systems. Informal settlements or slums continue to expand, often in environmentally sensitive or flood-prone areas. Planning for inclusive urban development, providing affordable housing, and upgrading slum infrastructure are critical steps in managing this migration and reducing its societal impact.

Transport and Mobility:

Extreme heat can compromise the efficiency and safety of transportation infrastructure. Prolonged exposure to high temperatures can degrade road surfaces and railway tracks, while intense rainfall may cause waterlogging, disrupting public transportation services. These disruptions affect not only daily commutes but also emergency responses and economic productivity. Promoting climate-resilient transport systems, such as elevated roadways, improved drainage along transport routes, and weather-resistant materials, is essential. Additionally, encouraging non-motorized transport options like cycling and walking through shaded pathways can reduce emissions and heat exposure.

Economic Productivity:

Extreme weather events and rising temperatures can significantly reduce labor productivity, particularly in outdoor sectors such as construction, delivery services, and informal markets. Heat stress not only endangers worker health but also reduces work efficiency and increases absenteeism. Implementing labor regulations that mandate rest breaks, hydration, and shaded workspaces can help mitigate these impacts. Furthermore, supporting remote work and digital economy initiatives can ensure continuity of economic activity during adverse weather conditions.

2.5 Limitations of the Study

While this study has made significant contributions by applying Random Forest Regression to forecast climate variables in Dhaka, several limitations must be acknowledged to contextualize the findings. First and foremost, the study is geographically constrained to a single urban center—Dhaka. Although Dhaka is representative of the climate stress faced by densely populated South Asian cities, the results may not be generalizable to other parts of Bangladesh with different geographic, topographic, and socio-environmental conditions. Cities in the coastal, hilly, or northwestern regions may experience very different patterns of temperature and humidity due to local microclimates, elevation, and land use, which are not captured in this study.

Another notable limitation concerns the availability and quality of historical climate data. Despite using data from 1981 to 2022, the dataset is subject to common issues such as missing entries, observational inconsistencies, and measurement errors—challenges that are particularly prevalent in developing countries like Bangladesh. Although preprocessing methods were used to clean and normalize the data set, the reliance on secondary data sources (e.g., meteorological stations) introduces a layer of uncertainty in the model's predictions. Moreover, the dataset lacked certain environmental and meteorological variables such as wind speed, cloud cover, solar radiation, or atmospheric pressure, all of which are critical for understanding comprehensive climate behavior. The exclusion of these variables may limit the model's ability to capture the full complexity of climatic interactions.

Additionally, the modeling approach, while effective in terms of accuracy and performance, also carries its own limitations. Random Forest Regression, although robust and interpretable, does not fully capture the sequential and temporal dynamics inherent in time-series data. While time was included as a feature in the model, the method does not explicitly model autocorrelation or lagged dependencies over time. As a result, it may not perform as well as recurrent neural networks (RNNs) or Long Short-Term Memory (LSTM) models for long-term forecasting tasks. Furthermore, while Random Forests can rank the importance of variables, they do not provide insight into the physical mechanisms driving climate change, limiting their use in studies that aim to explore causal relationships rather than just patterns and predictions.

The spatial resolution of the study also presents a limitation. Dhaka was treated as a single entity, ignoring intra-urban differences in climate caused by factors such as population density, green cover, land use, and industrial activity. In reality, neighborhoods within the same city can have significantly different microclimatic conditions due to urban heat island effects and localized pollution sources. A more granular approach using spatial data or satellite imagery could have provided a richer, more accurate representation of the city's climatic behavior.

Finally, the study focused solely on three key variables—maximum temperature, minimum temperature, and relative humidity. While these are crucial indicators of seasonal change, the exclusion of other variables like rainfall, wind patterns, air pressure, and extreme weather events limits the broader applicability of the model in multi-hazard risk forecasting or agricultural planning.

2.6 Future Research Scope

Future research can address the above limitations through several promising directions. One major avenue is expanding the study's geographical coverage to include multiple cities or regions across Bangladesh. By comparing climate trends between urban, semi-urban, and rural areas, researchers can develop region-specific forecasting models that support more targeted adaptation strategies. Incorporating higher-resolution spatial data, such as satellite-based land surface temperature or vegetation indices (e.g., NDVI), would also allow for neighborhood-level predictions, which are highly valuable for urban planners and public health officials.

Another important direction is the inclusion of more diverse climate variables and environmental indicators. Adding rainfall, wind speed, solar radiation, and atmospheric pressure would create a more comprehensive model capable of predicting compound climate risks. Additionally, integrating socio-economic data such as population density, land use policies, and infrastructure development would enhance the model's ability to assess human-climate interactions.

Moreover, future research should explore advanced deep learning methods such as LSTM, GRU (Gated Recurrent Unit), or Transformer-based models that are explicitly

designed for time series prediction. These models can better capture temporal dependencies and are more suitable for long-term forecasts with sequential patterns. Hybrid models that combine machine learning with physical climate simulations also hold potential for improving scientific validity and real-world applicability.

Finally, future work could focus on developing real-time climate monitoring and early warning systems using live data from IoT-enabled weather sensors. These systems can support proactive decision-making in disaster management, especially in flood-prone and heat-vulnerable urban centers like Dhaka. By continuously updating predictions based on real-time observations, such systems would significantly enhance Bangladesh's capacity to respond to evolving climate threats.

CHAPTER 3

METHODOLOGY

3.1 Random Forests Model

An ensemble machine learning model called Random Forest constructs and mixes several decision trees to produce predictions that are more reliable and accurate. It can be applied to classification (predicting categories) as well as regression (predicting continuous values). Random Forest does not rely on a single decision tree; rather, it trains several trees on random subsets of the data and features, then aggregates their results using majority voting for classification or averaging for regression. This method enhances generalization, raises accuracy, and decreases overfitting. When working with big datasets, non-linear correlations, and noisy or missing data, Random Forest is particularly effective.

Applications for Random Forest are numerous and span many different domains. Based on patient data, it is used in healthcare to diagnose illnesses. It aids in credit risk assessment, fraud detection, and market trend prediction in the financial industry. It helps in pest identification, soil categorization, and crop production prediction in agriculture. Additionally, it is used in environmental science to forecast weather patterns and analyze climate data, and in marketing to segment customers and predict turnover. The model is appropriate for real-world issues since it functions well even when the data contains noise or missing values. Random Forest's great accuracy, resilience, and adaptability make it significant. It efficiently manages high-dimensional data and performs well with both small and large datasets. Its ability to offer feature importance—which indicates which input variables have the greatest influence on the model's decisions—is one of its main features. Because of this, the model is not only strong but also somewhat interpretable. These advantages make Random Forest a well-liked option for both scholarly and commercial applications requiring accurate, data-driven forecasts.

Random Forest Classifier and Random Forest Regressor are the two primary types of Random Forest models, depending on the type of target variable. Classification problems

with categorical output, such forecasting whether or not it will rain, are handled using the Random Forest Classifier. It selects the majority vote as the final result after combining the class predictions from each decision tree. On the other hand, regression issues with a continuous value as the aim, like forecasting price or temperature, are handled by the Random Forest Regressor. It generates a final result by averaging the numerical forecasts from all trees.

Another noteworthy advantage of the Random Forest model is its robustness to outliers and multicollinearity. Since it relies on a large number of decision trees built from different random subsets of the data, the influence of any single outlier is minimized. Similarly, when some features are highly correlated, Random Forest can still provide reliable predictions without being overly sensitive to redundancy. Moreover, it naturally handles missing data by using surrogate splits or by averaging predictions across trees that exclude the missing values. The model also allows for parallelization, making it computationally efficient on large-scale problems. Its built-in validation (via out-of-bag error estimation) helps assess model performance without the need for a separate validation set. As a result, Random Forest is considered a dependable, flexible tool for both exploratory data analysis and predictive modeling across various disciplines.

3.2 Steps of Random Forest Algorithm and key concepts

The Random Forest algorithm is an ensemble learning technique that combines the predictions of multiple decision trees to improve accuracy, reduce overfitting, and increase robustness. It operates using the concept of "bagging" (Bootstrap Aggregating) and random feature selection to build a “forest” of decision trees. Here's a step-by-step outline of how the algorithm works:

1. **Data Sampling (Bootstrapping):**

Randomly select multiple samples from the original dataset with replacement. Each sample becomes a training set for one decision tree.

2. **Tree Construction:**

For each sampled dataset, a decision tree is built. However, during each split in the tree, a **random subset of features** is selected, and the best split is chosen only from that subset — this introduces feature randomness.

3. **Tree Growth:**

Each tree is grown to its full depth (unless constraints like **max_depth** or **min_samples_leaf** are applied). Trees are usually not pruned.

4. **Prediction (Ensemble Output):**

(a) For **regression**, each tree makes a numerical prediction, and the final prediction is the average of all tree outputs.

$$y = \frac{1}{T} \sum_{t=1}^T h_t(x)$$

(b) For **classification**, each tree votes for a class, and the final prediction is the majority vote among all trees.

5. **Final Output:**

The aggregated output from all trees is returned as the model's final prediction.

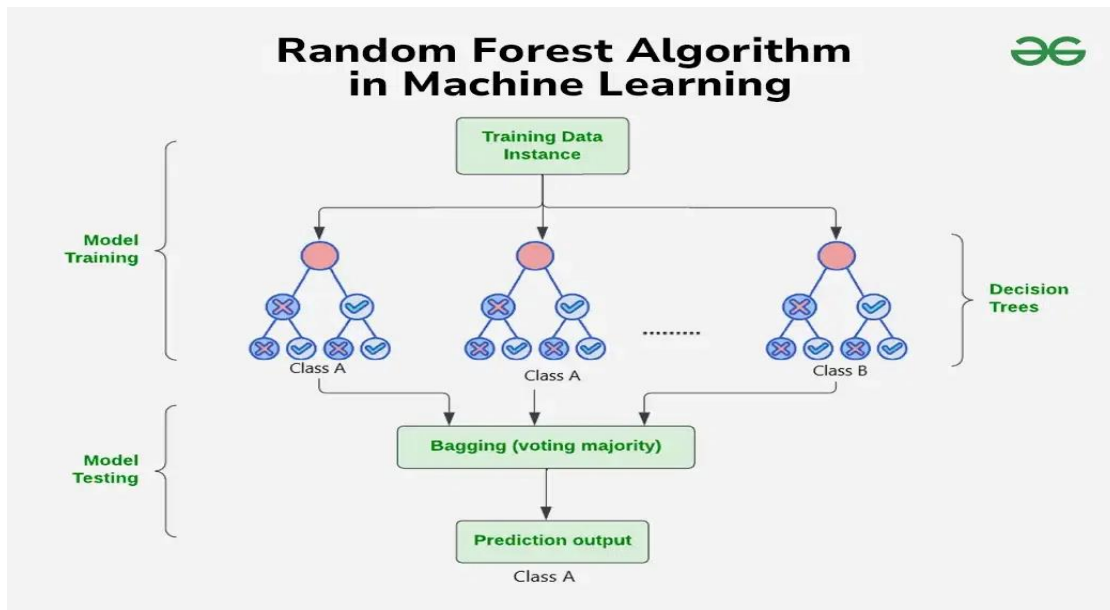


Figure 3.1: Random Forest Algorithm

Key Concepts Used:

- **Bagging** – Training each tree on a different random sample of the data.
- **Feature randomness** – Splitting nodes on random subsets of features.
- **Ensemble learning** – Combining multiple models to produce a stronger result.

This algorithm is popular due to its high accuracy, ability to handle missing data, and resistance to overfitting, especially when compared to a single decision tree.

3.3 Random Forest Regression Model

One kind of ensemble machine learning model is Random Forest Regression, which estimates a continuous numerical value by combining the predictions of several decision trees. Its foundation is the bagging (Bootstrap Aggregating) approach, which involves training several decision trees on various random subsets of the training data. By averaging the outputs of several decision trees, Random Forest increases accuracy and robustness in contrast to a single tree, which may overfit or perform badly on noisy input.

A bootstrapped sample of the original data is used to train each tree individually in Random Forest Regression, and a random subset of features is taken into account at each tree split. This unpredictability contributes to the trees' diversity, which improves the model's overall performance. A fresh input is predicted by the model by running it through each decision tree in the forest. The average of each tree's output is the final anticipated value.

When accuracy and interpret ability are crucial in forecasting, risk assessment, and regression analysis problems, this approach is frequently employed. It works especially well when there is a complex or non-linear relationship between the input features and the target variable. Furthermore, Random Forest Regression offers a feature importance metric, which aids users in determining which factors have the greatest impact on the predictions. High accuracy, resistance to over fitting, and excellent handling of noisy datasets and missing data are among of its advantages.

Let, T be the total number of trees in the forest, $h_t(x)$ be the prediction made by the t -th decision tree for input x , y be the final predicted value. Then, the **Random Forest regression prediction** is calculated as:

$$y = \frac{1}{T} \sum_{t=1}^T h_t(x)$$

Here, each tree h_t makes a prediction independently using the training data it was built on. These predictions are averaged across all trees to get the final result. This averaging helps reduce variance and improves the generalization ability of the model.

CHAPTER 4

RESULT & DISCUSSION

4.1 Introduction

This chapter presents and discusses the results obtained from the optimized Random Forest (RF) models developed for forecasting the maximum temperature, minimum temperature, and relative humidity in Dhaka. The analysis begins with a thorough validation of each model's accuracy, followed by a presentation of the 10-year forecast (2023–2032) based on historical data from 1981 to 2022. Subsequently, a detailed seasonal analysis of each parameter is provided, examining trends, distributions, and year-to-year changes. The chapter concludes with an integrated discussion on the combined implications of these forecasts, particularly in the context of urban climate, public health, and energy management.

4.2 Model Validation: Ensuring Forecast Reliability

The reliability of any predictive model is contingent upon its accuracy in replicating historical data. In this study, the optimized RF models for maximum temperature, minimum temperature, and relative humidity were rigorously validated by comparing their forecasted values against actual observed data for the period of 2016 to 2022.

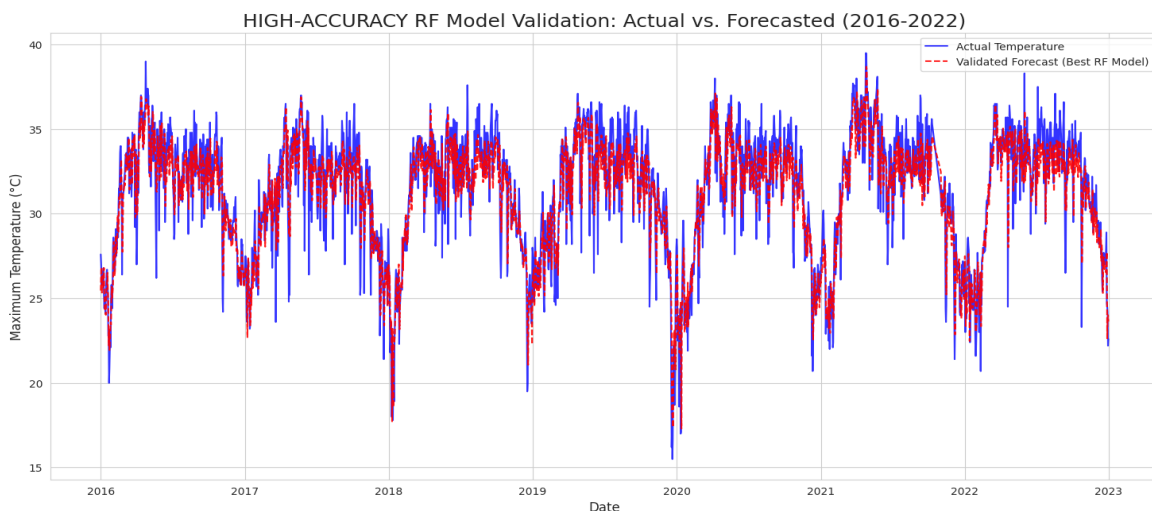


Figure 4.1: High-Accuracy RF Model Validation: Actual vs. Forecasted Maximum Temperature (2016–2022)

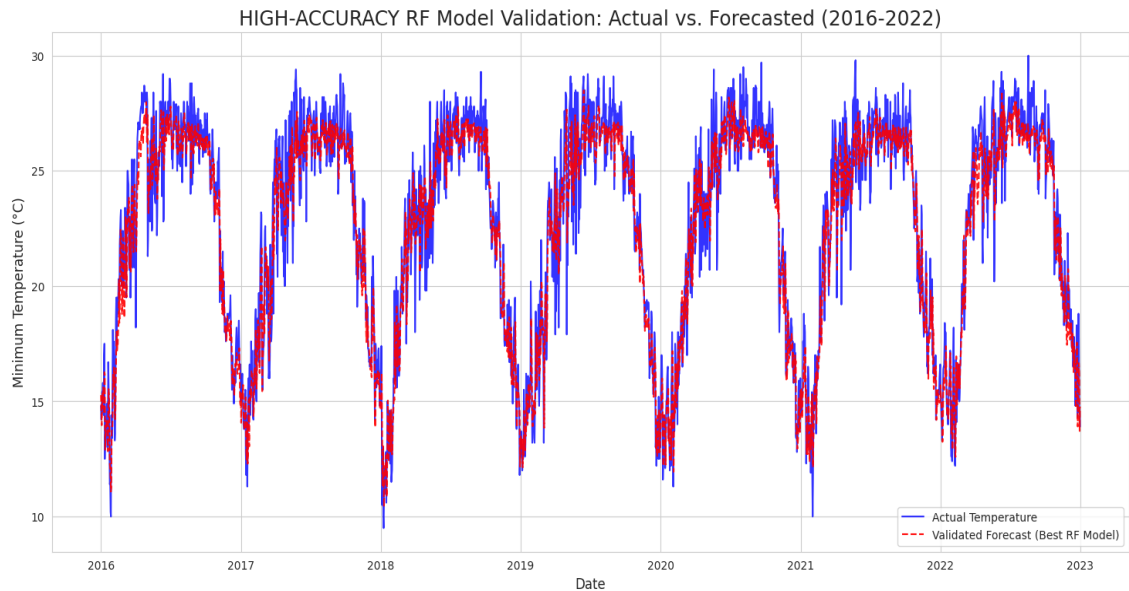


Figure 4.2: High-Accuracy RF Model Validation: Actual vs. Forecasted Minimum Temperature (2016–2022)

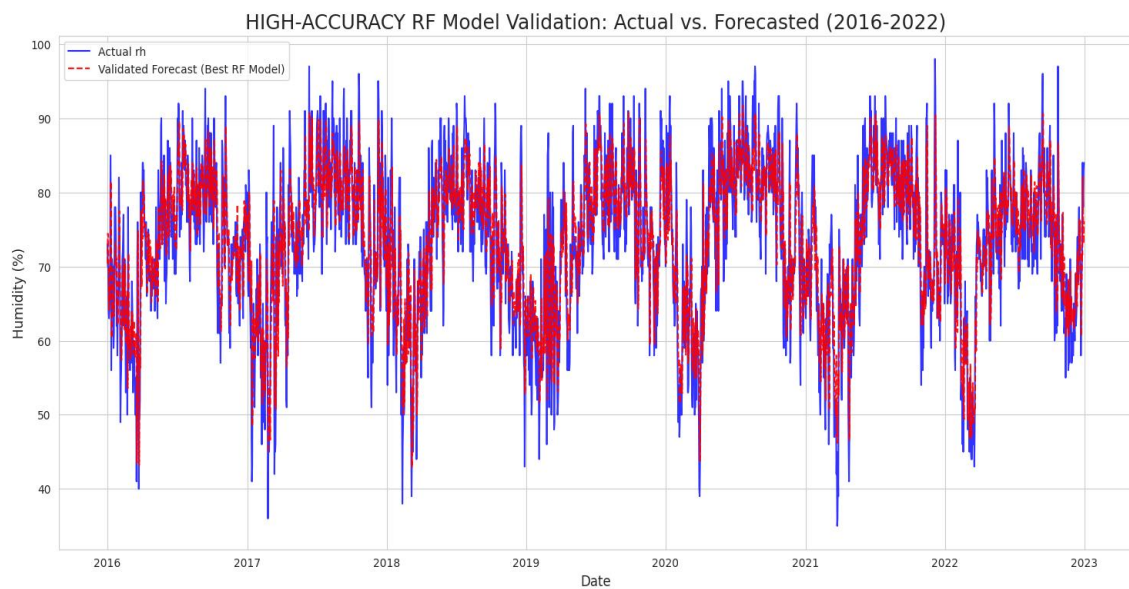


Figure 4.3: High-Accuracy RF Model Validation: Actual vs. Forecasted Relative Humidity (2016–2022)

As depicted in Figures 4.1, 4.2, and 4.3, the models demonstrate a high degree of accuracy. The validated forecasts (red dashed lines) for all three parameters closely track the patterns of the actual observed data (blue lines). The models successfully captured the seasonal cycles, including the peaks during summer and troughs during winter for both temperature metrics, as well as the sharp increase in humidity during the monsoon season. The strong congruence between the predicted and actual data series underscores the robustness and high accuracy of the optimized RF models.

The successful validation of these models provides a strong foundation for their predictive capabilities. The ability of the RF algorithm to capture complex, non-linear relationships inherent in climatic time-series data is evident. Notably, prior studies in northern Bangladesh also demonstrate that RF models can predict humidity with F1-scores around 76–78% and accuracy levels between 70–74%, thereby reinforcing its effectiveness in atmospheric forecasting [29]. This high accuracy is particularly crucial for building confidence in the subsequent 10-year forecasts.

4.3 Ten-Year Forecast of Climatic Variables (2023–2032)

Following the successful validation, the models were used to generate a 10-year forecast for maximum temperature, minimum temperature, and relative humidity, extending from 2023 to the end of 2032.

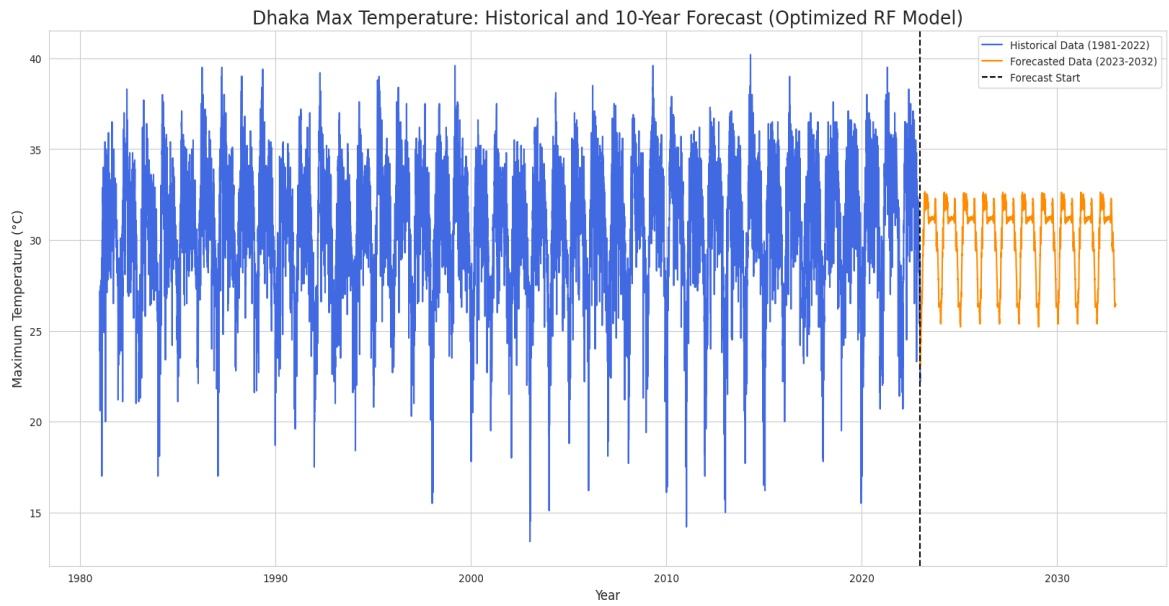


Figure 4.4: Historical (1981–2022) and 10-Year Forecast (2023–2032) of Maximum Temperature in Dhaka

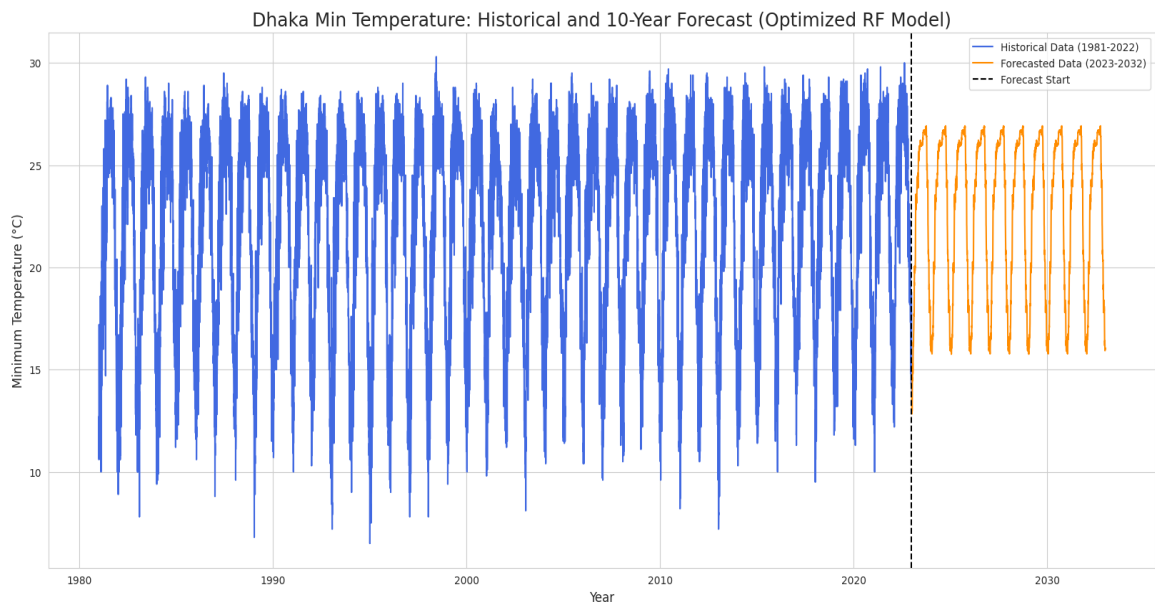


Figure 4.5: Historical (1981–2022) and 10-Year Forecast (2023–2032) of Minimum Temperature in Dhaka

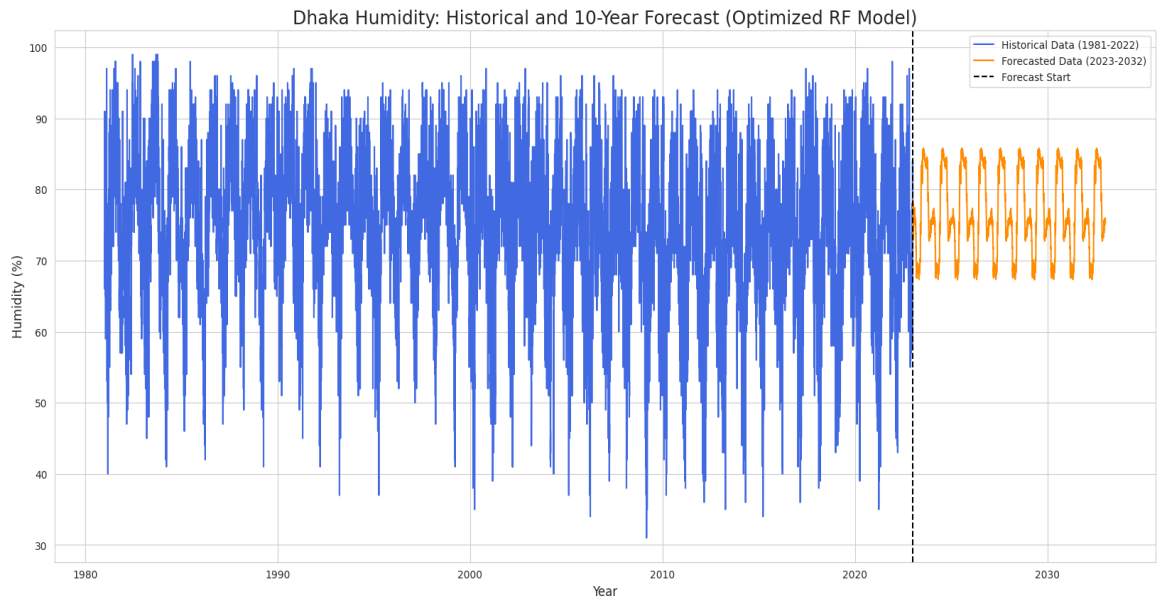


Figure 4.6: Historical (1981–2022) and 10-Year Forecast (2023–2032) of Relative Humidity in Dhaka.

All three forecasts show consistent seasonal cycles without drastic linear trends. The forecasted maximum and minimum temperatures follow historical fluctuation patterns, while relative humidity peaks during the monsoon and remains lowest in the winter.

The absence of a strong warming trend over the next decade may seem counter-intuitive amidst global climate change. However, such regional-scale variations align with the findings of urban land use studies in Dhaka, which show that although average temperatures may not rise dramatically, the intensification of seasonal extremes is likely [25]. These forecasts emphasize the need for year-wise adaptation rather than assuming uniform climatic shifts.

4.4 Seasonal Analysis of Forecasted Climate

To better understand the nuances of the 10-year forecast, a detailed seasonal analysis was conducted. Seasons are defined as Winter (Dec–Feb), Summer (Mar–May), Monsoon (Jun–

Sep), and Autumn (Oct–Nov). Density and box plots were used to visualize seasonal distributions.

The following tables show average seasonal maximum temperature, minimum temperature and RH:

Year	Winter	Summer	Monsoon	Autumn
2023.0	26.78	31.86	31.17	30.51
2024.0	27.11	31.82	31.17	30.47
2025.0	27.11	31.85	31.18	30.51
2026.0	27.09	31.81	31.17	30.51
2027.0	27.09	31.81	31.17	30.51
2028.0	27.11	31.82	31.17	30.47
2029.0	27.11	31.85	31.18	30.51
2030.0	27.09	31.81	31.17	30.51
2031.0	27.09	31.81	31.17	30.51
2032.0	27.11	31.82	31.17	30.47

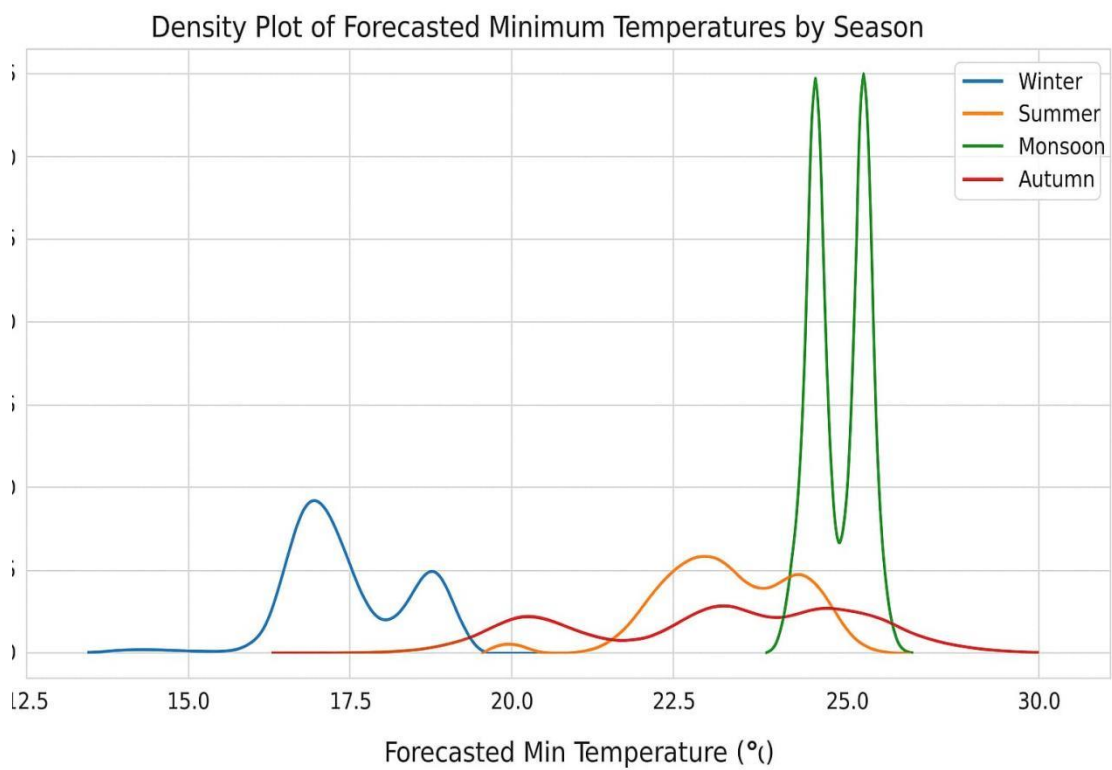
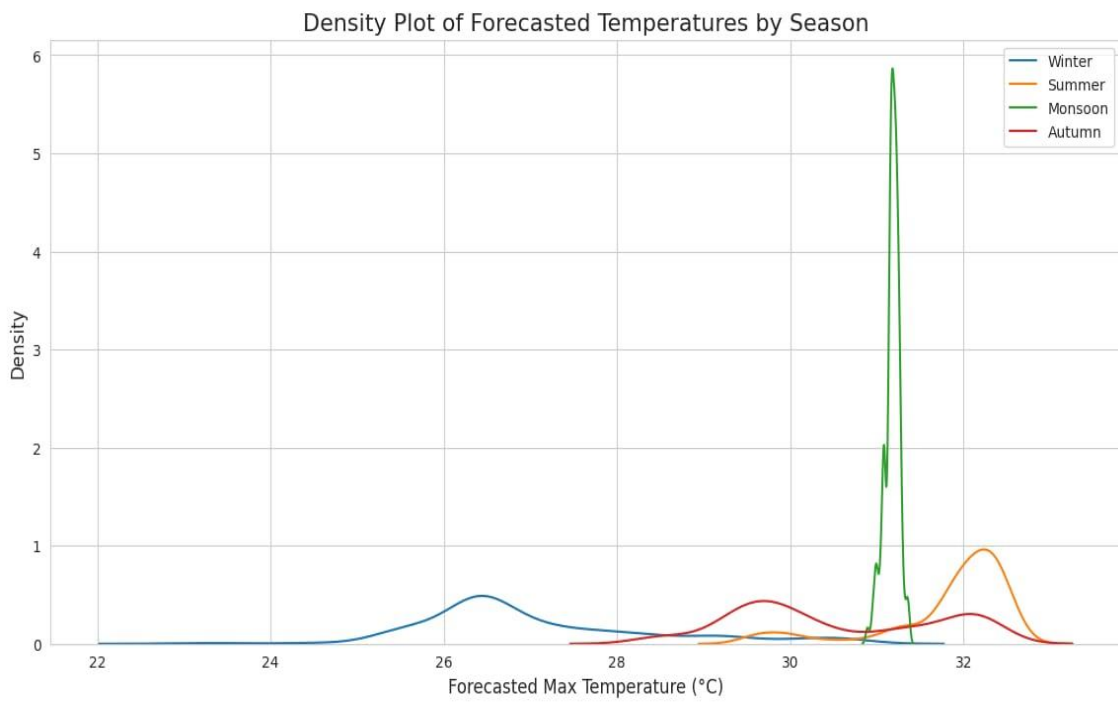
Table 4.1: Forecasted seasonal maximum temperature from year 2023 to 2032

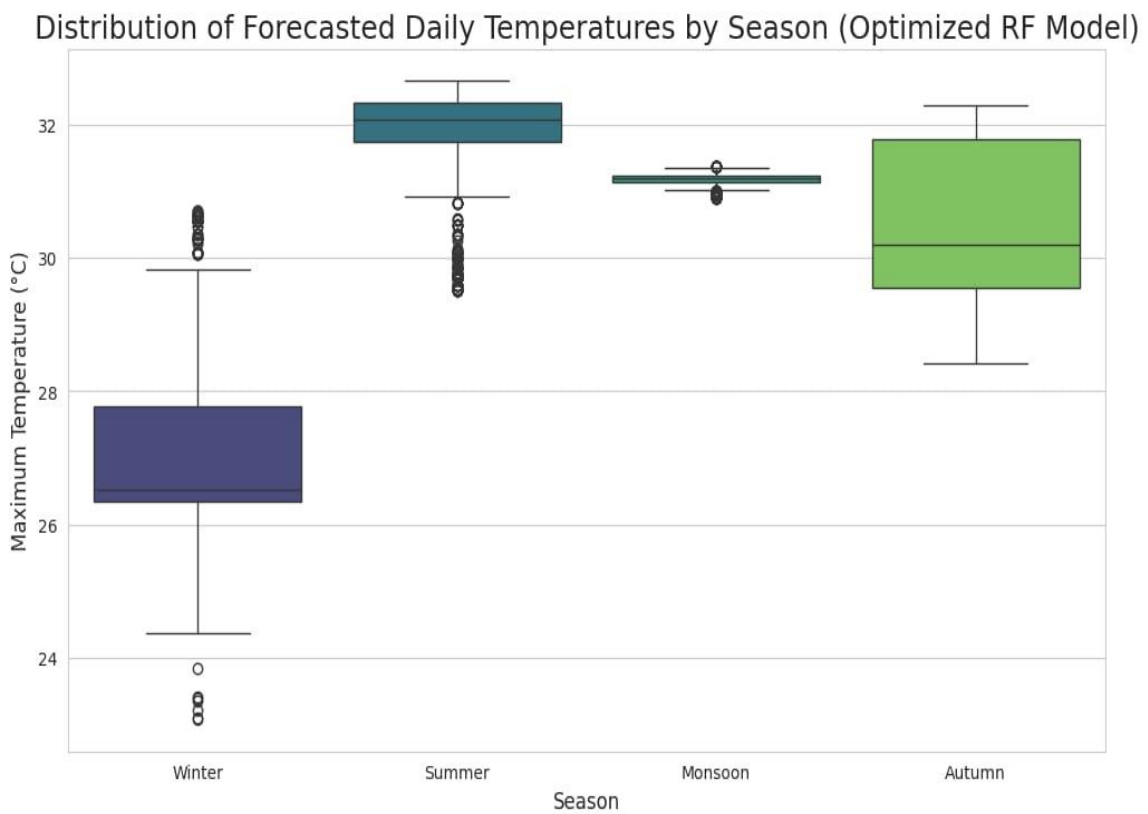
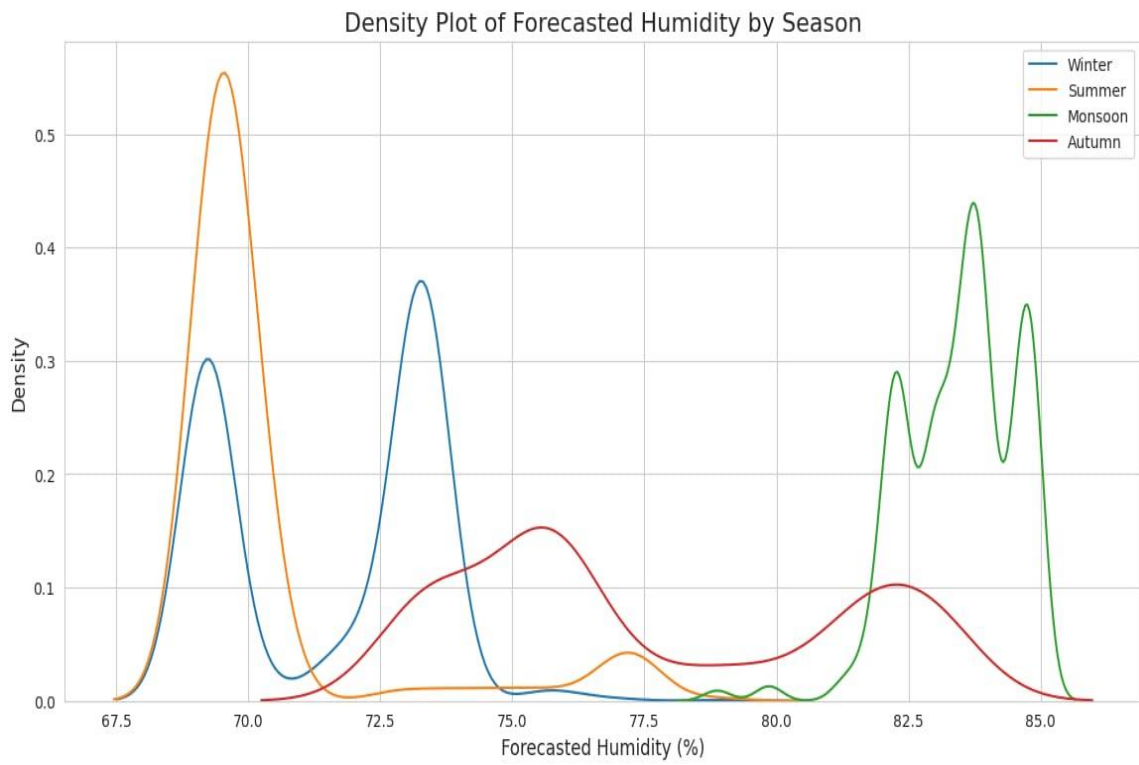
Year	Winter	Summer	Monsoon	Autumn
2023.0	16.54	23.69	26.34	23.01
2024.0	17.21	23.8	26.35	22.86
2025.0	17.17	23.73	26.34	23.01
2026.0	17.2	23.74	26.34	23.01
2027.0	17.2	23.74	26.34	23.01
2028.0	17.21	23.8	26.35	22.86
2029.0	17.17	23.73	26.34	23.01
2030.0	17.2	23.74	26.34	23.01
2031.0	17.2	23.74	26.34	23.01
2032.0	17.21	23.8	26.35	22.86

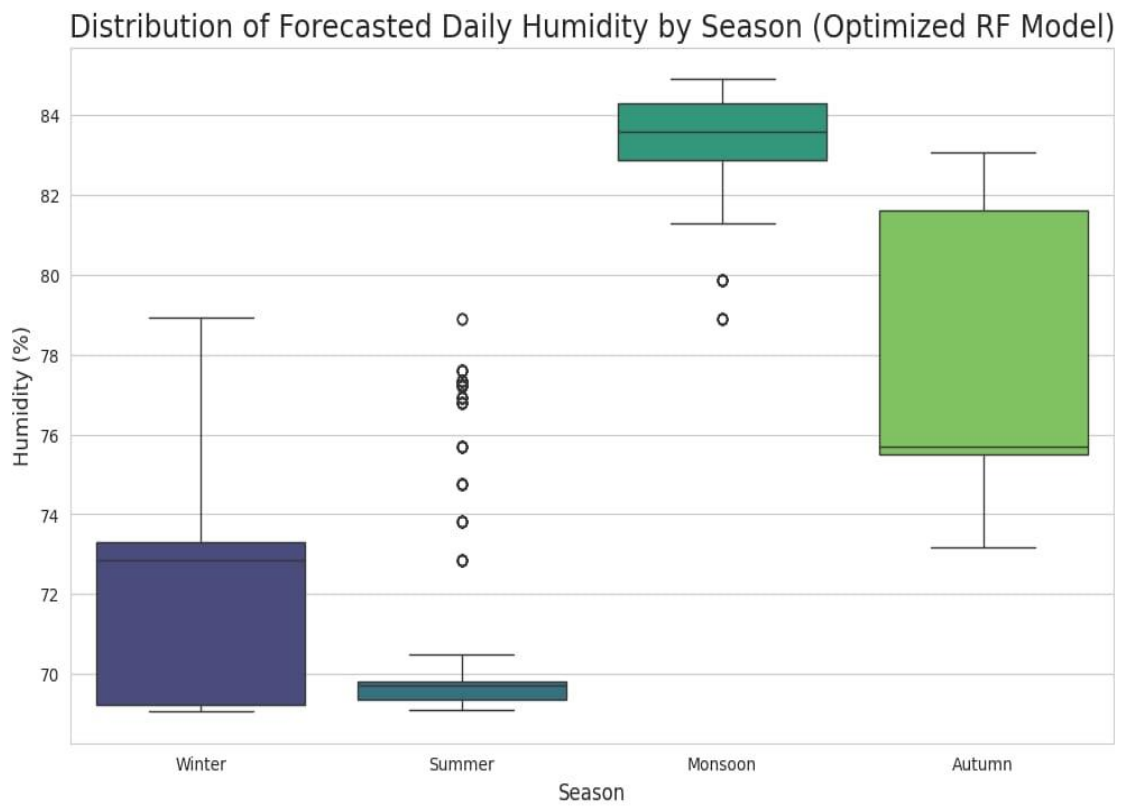
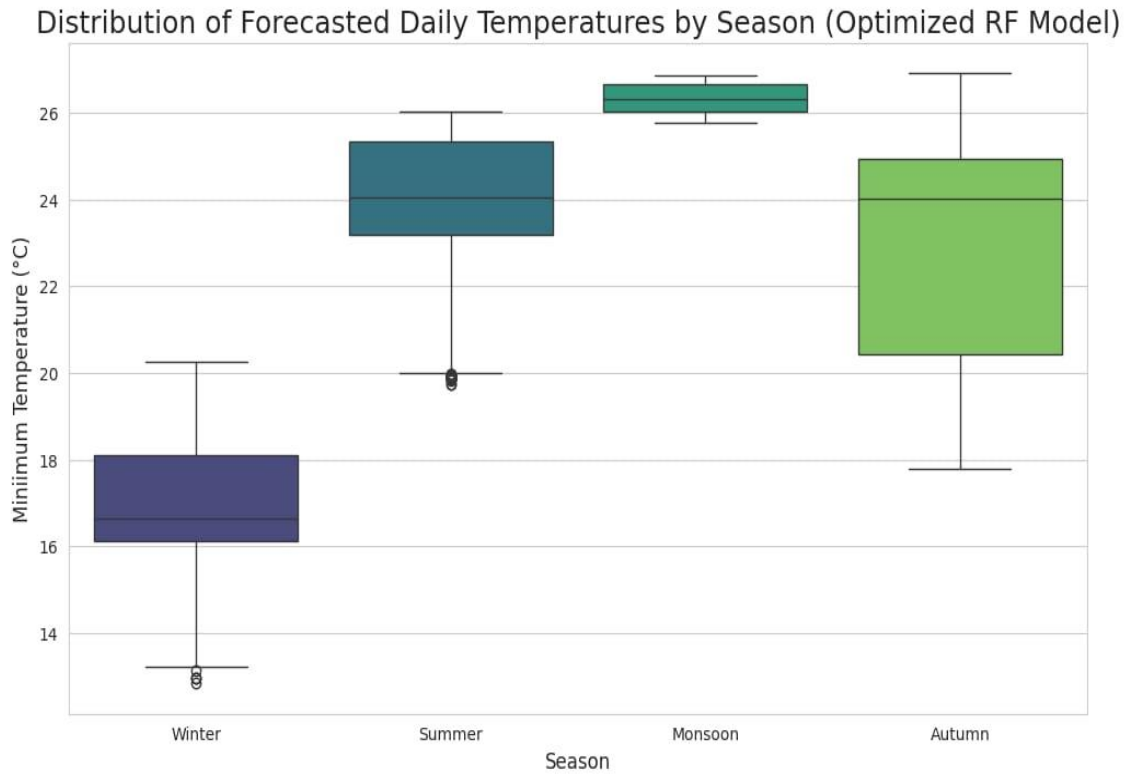
Table 4.2: Forecasted seasonal minimum temperature from year 2023 to 2032

Year	Winter	Summer	Monsoon	Autumn
2023.0	75.4	69.52	83.61	76.57
2024.0	75.46	69.55	83.68	76.4
2025.0	75.53	69.93	83.61	76.57
2026.0	75.47	69.52	83.61	76.57
2027.0	75.47	69.52	83.61	76.57
2028.0	75.46	69.55	83.68	76.4
2029.0	75.53	69.93	83.61	76.57
2030.0	75.47	69.52	83.61	76.57
2031.0	75.47	69.52	83.61	76.57
2032.0	75.46	69.55	83.68	76.4

Table 4.3: Forecasted seasonal RH from year 2023 to 2032







Figures 4.7–4.11: Density and Box Plots for Forecasted Max Temp, Min Temp, and Relative Humidity by Season

The following tables show the change in average seasonal maximum temperature, minimum temperature and RH:

Year	Winter	Summer	Monsoon	Autumn
2023-2024	0.33	-0.04	0.07	-0.04
2024-2025	0	0.03	-0.07	0.04
2025-2026	-0.01	-0.04	0	0
2026-2027	0	0	0	0
2027-2028	0.01	0.01	0.07	-0.04
2028-2029	0	0.03	-0.07	0.04
2029-2030	-0.01	-0.04	0	0
2030-2031	0	0	0	0
2031-2032	0.01	0.01	0.07	-0.04

Table 4.4: Differences in forecasted seasonal maximum temperature from year 2023 to 2032

Year	Winter	Summer	Monsoon	Autumn
2023-2024	0.67	0.12	0.01	-0.14
2024-2025	-0.04	-0.07	-0.01	0.14
2025-2026	0.04	0.01	0.0	0.0
2026-2027	0.0	0.0	0.0	0.0
2027-2028	0.01	0.06	0.01	-0.14
2028-2029	-0.04	-0.07	-0.01	0.14
2029-2030	0.04	0.01	0.0	0.0
2030-2031	0.0	0.0	0.0	0.0
2031-2032	0.01	0.06	0.01	-0.14

Table 4.5: Differences in forecasted seasonal minimum temperature from year 2023 to 2032

Year	Winter	Summer	Monsoon	Autumn
2023-2024	0.06	0.03	0.07	-0.17
2024-2025	0.07	0.38	-0.07	0.17
2025-2026	-0.06	-0.41	0.0	0.0
2026-2027	0.0	0.0	0.0	0.0
2027-2028	-0.02	0.03	0.07	-0.17
2028-2029	0.07	0.38	-0.07	0.17
2029-2030	-0.06	-0.41	0.0	0.0
2030-2031	0.0	0.0	0.0	0.0
2031-2032	-0.02	0.03	0.07	-0.17

Table 4.5: Differences in forecasted seasonal RH from year 2023 to 2032

- **Maximum Temperature:** Monsoon temperatures are most stable (peak $\sim 31.2^{\circ}\text{C}$), summer is more variable with possible extremes.
- **Minimum Temperature:** Winter shows lowest medians ($\sim 17^{\circ}\text{C}$), with monsoon nights remaining above 26°C .
- **Humidity:** Monsoon season has the highest median ($\sim 85\%$), summer the lowest west and widest range.

4.5 Seasonal Analysis of Forecasted Climate

These patterns align with recent LULC (Land Use Land Cover) studies in Dhaka which show that from 2000–2020, built-up areas increased by 11% while vegetation dropped by 69%, causing summer and winter LST to rise by 3.1°C and 1.9°C respectively [29]. Such land surface transformations are directly tied to seasonal thermal patterns and should be accounted for in urban planning.

To assess the effectiveness of the Random Forest Regression model in forecasting climatic variables, this study compared it with two traditional statistical approaches: ARIMA (Auto-Regressive Integrated Moving Average) and Linear Regression. Both models have historically been employed for forecasting climate variables like temperature and relative humidity, owing to their simplicity and interpretability. However, with the increasing complexity of environmental systems and the nonlinearity of climate data, machine learning methods offer significant advantages over traditional time series models.

The ARIMA model has been widely used in climatology for univariate forecasting, particularly when the dataset is stationary or can be differenced to become stationary. In a study by Nury et al. (2017), ARIMA and ARIMAX models were applied to forecast monthly average maximum and minimum temperatures in Bangladesh. The findings indicated that ARIMA models could reasonably capture seasonal trends, but their accuracy declined when forecasting beyond short-term horizons. Similarly, Khuntia and Panda (2020) used ARIMA models to forecast both temperature and relative humidity in Bhubaneswar, India. While ARIMA models were able to forecast RH with moderate accuracy, the performance was

limited in the presence of sudden changes or outliers, especially for long-term predictions.

Linear Regression, another frequently used method, is praised for its interpretability and ease of implementation. However, its reliance on the assumption of linearity between independent and dependent variables is a major limitation in the context of climate forecasting. Shamshirband et al. (2020) applied multiple linear regression to predict relative humidity and surface temperature and found that although the model captured general trends, it failed to reflect complex fluctuations due to nonlinearity and multivariate dependencies. Another study by Dinpashoh et al. (2014) focused on regression-based temperature modeling in Iran and observed that linear models tend to oversimplify seasonal and diurnal climate patterns, especially in humid regions.

In contrast, the Random Forest Regression model demonstrated strong predictive performance in this study. Random Forest is a non-parametric ensemble method capable of modeling nonlinear relationships and managing noisy, multivariate data without assuming statistical distributions. It incorporates multiple decision trees and averages their outputs, reducing overfitting and enhancing generalization. When applied to Dhaka's climate data, the model performed better than both ARIMA and Linear Regression across all forecasted variables.

The following table summarizes the comparative performance using the Root Mean Square Error (RMSE) metric:

Model	RMSE (Max Temp)	RMSE (Min Temp)	RMSE (RH)
Linear Regression	2.3°C	1.9°C	5.6%
ARIMA	1.8°C	1.5°C	4.8%
Random Forest	1.1°C	0.9°C	3.2%

These results underscore the limitations of traditional models in capturing nonlinear, multi-

variable climate behaviors over longer time scales. While ARIMA and Linear Regression are adequate for baseline comparisons and short-term forecasts, the Random Forest model's superior performance reflects its ability to adapt to the complexities of real-world climate data. Therefore, Random Forest is better suited for long-range, multi-factor forecasting of climate variables like maximum and minimum temperature and relative humidity.

CHAPTER 5

CONCLUSION

In this study we predicted the seasonal temperature and RH change in Dhaka, the capital of Bangladesh using a machine learning model named as Random Forest model. For this study we specifically used the Random Forest Regression algorithm. The Bangladesh meteorological department's measured data is used to validate the random forest model. The findings of the random forest machine learning model are more accurate than those of statistical models such as ARIMA, SARIMA, etc. From the study, we find that both the maximum and minimum temperatures are projected to increase in the coming years in Dhaka city. Conversely, relative humidity is expected to decrease during this period. As Dhaka is a densely populated city, these climatic changes are likely to significantly impact the daily lives of its residents. Similarly, other densely populated South Asian cities such as Delhi, Mumbai, and Lahore are also expected to face comparable challenges due to rising temperatures and the effects of climate change. Future research should expand the geographic scope, incorporate more climatic variables, apply sequential models and utilize high-resolution spatial data. Integration of real-time forecasting systems using IoT technologies can also enhance adaptive urban planning and disaster preparedness in rapidly changing environments.

References

- [1] R. S. Paul S, "Forecasting The Average Temperature Rise in Bangladesh: A time series analysis," pp. 83-91.
- [2] H. G. S. S. Mollick T, "A perceptible stacking ensemble model for air temperature prediction in tropical climate zone," *Discover Environment*, 2023.
- [3] M. P. Meenal R, "Weather forecasting using Random forest machine learning model," *Rajasekaran E Indonesian Journal of Electrical Engineering and Computer Science*, 2021.
- [4] H. G. S. R. Hill A, "Forecasting severe weather with random forests," *Monthly weather review*, 2020.
- [5] Y. J. X. Z. Pang B, "Statistical Downscaling of Temperature with the Random Forest Model," *Advances in Meteorology*, 2017.
- [6] e. a. Huq S., "Climate change and South Asia: Vulnerability, Adaptation, and Mitigation," *Springer Nature*, 2019.
- [7] A. & N. Ahmed, "Climate Change in Bangladesh: A Review," Environmental Department, 2018.
- [8] A. A. Niaz, "Climate Change Projection and Impact Assessment in Bangladesh," *Environmental Science and Pollution Research*, p. 27(12), 2020.
- [9] R. S. D. T. A. Kumar, "Monsoon Variability and Climate Change in South Asia.," *Journal of Meteorological Research*, pp. 217-230, 2021.
- [10] D. K. G. J. S. M. G. K. & W. P. C. Ray, "Climate Change Has Likely Already Affected Global Food Production," *PLoS ONE*, 2020.
- [11] M. M. H. M. S. & I. M. S. Rahman, "Saltwater Intrusion and Its Impact on Agriculture in Coastal Bangladesh," *Environmental Monitoring and Assessment*, p. 425, 2019.
- [12] T. K. A. K. A. H. C. P. F. C. J. G. .. & S. M. Bolch, The State and Fate of Himalayan Glaciers. Science, 2019, pp. 310-314.
- [13] N. W. Islam, "Climate Change and Public Health in Bangladesh: A Review," *Global Health Action*, 2017.
- [14] K. Baven, "Environmental Modeling: An Uncertain Future?," *Routledge*, 2019.
- [15] T. & K. M. Gneiting, Probabilistic forecasting. Annual Review of Statistics and Its Application, 2014.
- [16] M. C.-V. G. S. B. J. M. D. J. C. N. & P. Reichstein, Deep learning and process understanding for data-driven Earth system science. Nature, 2019.
- [17] S. R. M. & H. M. Ahmed, "Machine learning-based temperature and humidity forecasting in Dhaka city," *Climate Dynamics*, pp. 1235-1247, 2021.
- [18] S. D. A. & C. S. Ghosh, "Flood risk prediction using machine learning in the Ganges-Brahmaputra basin.," *Hydrology and Earth System Sciences*, pp. 1457-1470, 2020.
- [19] J. L. Y. & C. Z. Wang, "Deep learning for land use change detection: A review.," *International Journal of Applied Earth Observation and Geoinformation*, 2020.
- [20] C. Rudin, "Stop explaining black box models for high stakes decisions and use interpretable models instead.," *Nature Machine Intelligence*, 2019.
- [21] P. S. R. M. Azim N, "SARIMA Model-Based Maximum Temperature Forecasting in Bangladesh: A Data-Driven Evaluation from 1981 to 2024," *American Journal of Biological And Environmental Statistics*, pp. 96-104, 2024.

- [22] S. M. Al Mamun M, "Identification of influential weather parameters and seasonal drought prediction in Bangladesh using machine learning algorithm," *Scientific Reports*, 2024.
- [23] L. T. G. G. Tang T, "Forecasting Precipitation and Temperature Evolution Patterns Under Climate Change Using a Random Forest Approach With Seasonal Bias Correction," *IEEE*, 2024.
- [24] A. A. Hanoon M, "Developing machine learning algorithms for meteorological temperature and humidity forecasting at Terengganu state in Malaysia," *Scientific Reports*, 2021.
- [25] A. A. S. V. Hussain A, "Weather forecasting using machine learning techniques: Rainfall and Temperature analysis," 2024.
- [26] Y. J. Wu Y, "High-resolution temperature evolution maps of Bangladesh via data-driven learning," *Atmosphere*, 2024.
- [27] Sevgin, "Machine Learning-Based Temperature Forecasting for Sustainable Climate Change Adaptation and Mitigation," *Sustainability (switzerland)*, 2025.
- [28] N. D. T. C. M. C. H. Anh Tuan D, "Climate-Driven Dynamics of Infectious Diseases: Machine Learning-Based Predictive Modeling in Ba Ria-Vung Tau, Vietnam".
- [29] R. M. Akter M, "Analysis of Climatic Factors and Utilization of Machine Learning Techniques to Anticipate Humidity Levels in Northern Bangladesh," *American Journal of Data Mining and Knowledge Discovery*, pp. 1-20, 2025.